

Friday, August 23, 2024 10:32 AM

PROBLEM 2

Let L_1 and L_2 be the following languages on the alphabet $\Sigma = \{0,1\}$:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

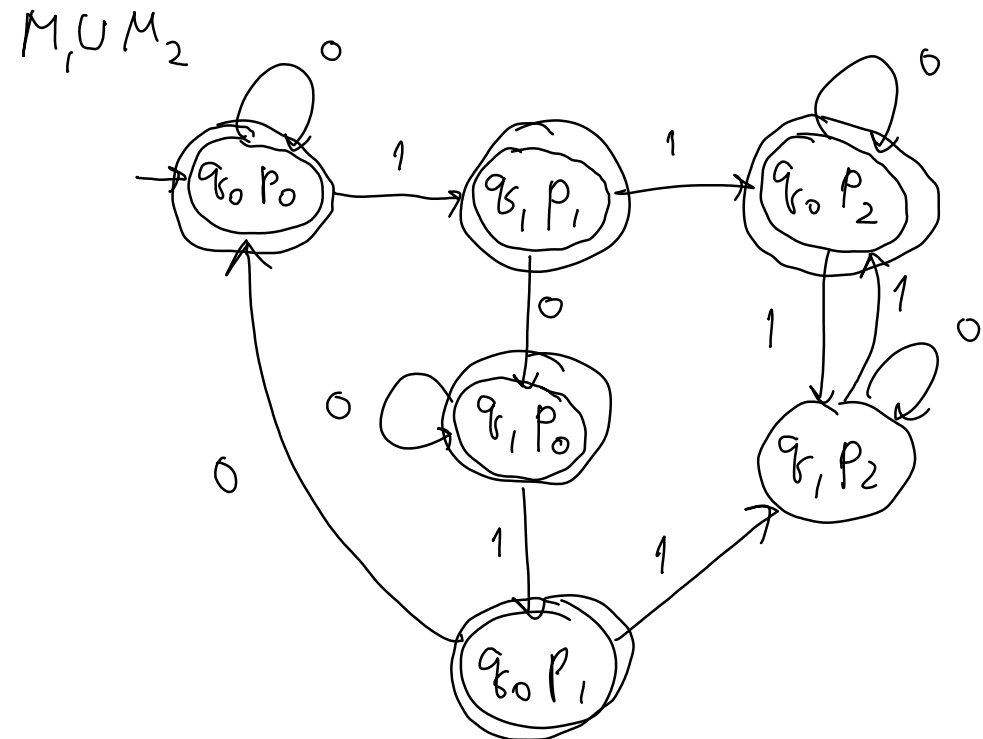
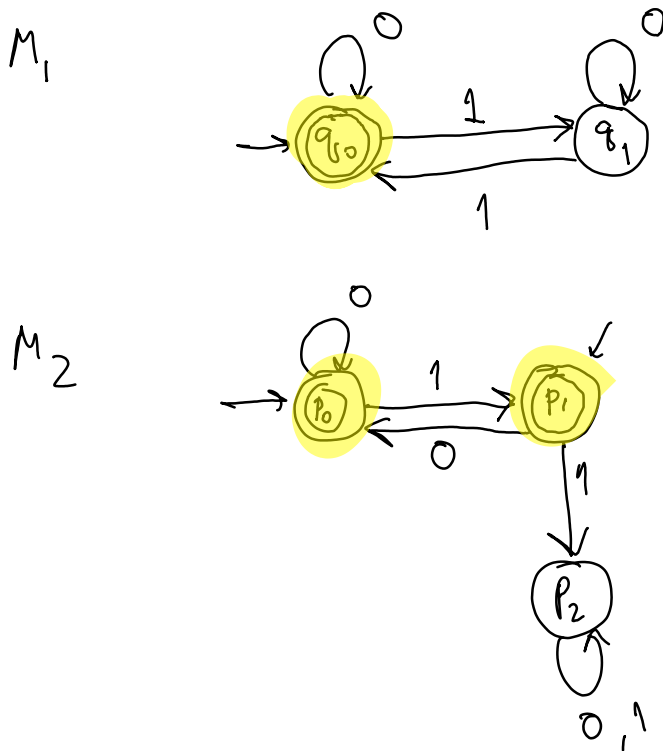
$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

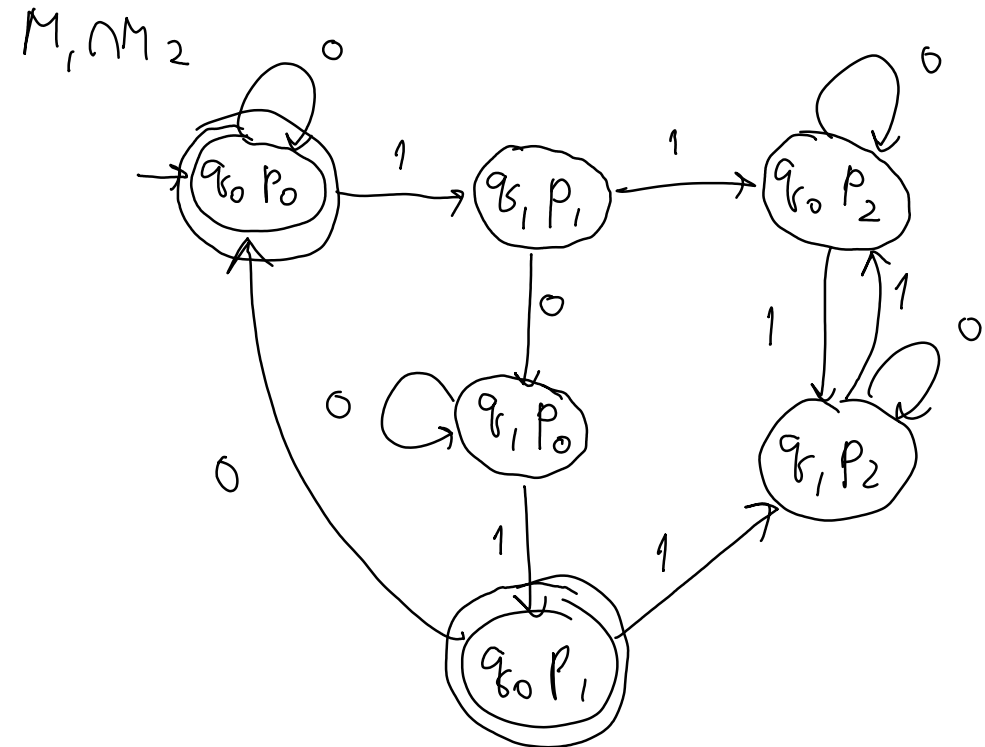
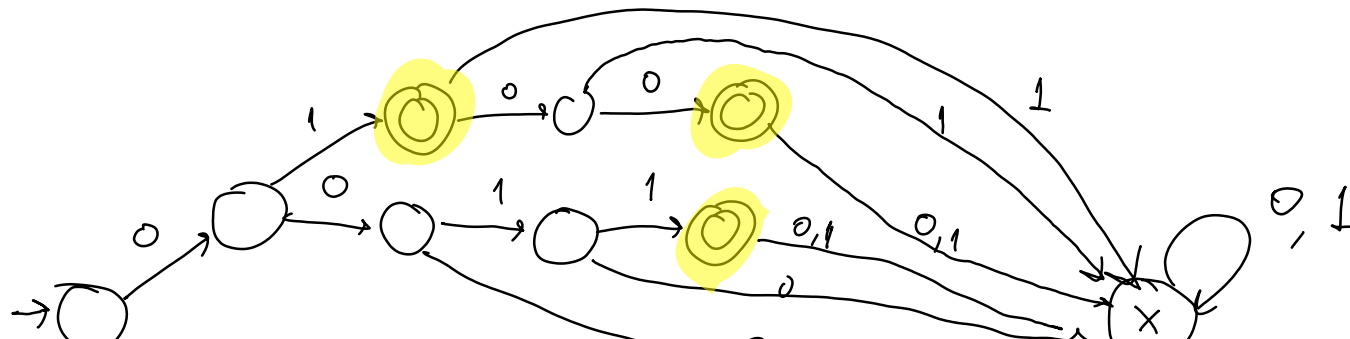
0 is even.

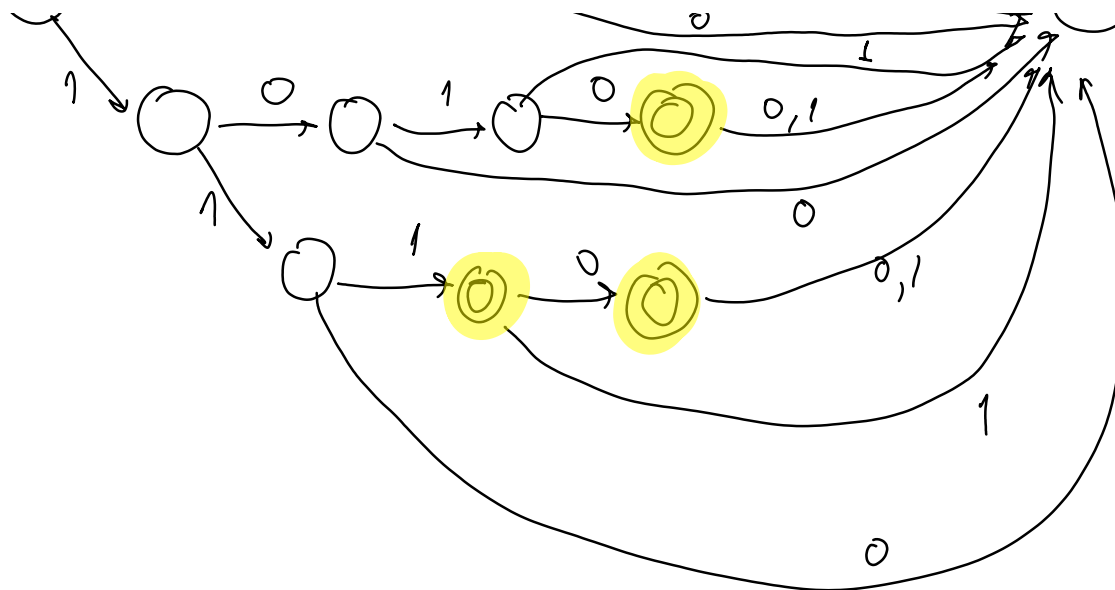
2.1 Design (deterministic) finite automata M_1 and M_2 that recognize L_1 and L_2 , respectively.

2.2 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cap L_2$.

2.3 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cup L_2$.




$$L = \{0011, 1010, 1110, 0100, 111, 01\}.$$




3.2 Design a (deterministic) finite automaton that recognizes the following language L' on the alphabet $\Sigma = \{0,1\}$:

$L' = \{w \in \Sigma^* \mid w \text{ is a binary representation of a non-negative integer divisible by 3}\}.$

0, 00, 000

1, 011, 0011

110, 0110

1001

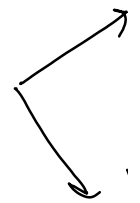
ε

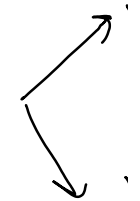
L_0 contains binary representations of non-negative integers divisible by 3

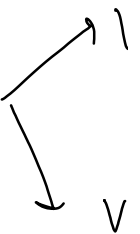
L_1 contains binary representations of non-negative integers that give remainder 1 when divided by 3

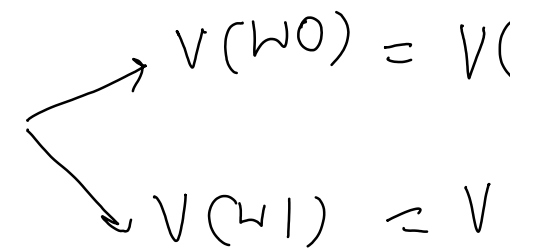
L_2 contains binary representations of non-negative integers that give remainder 2 when divided by 3

1100

$$w \in L_0 \rightarrow V(w) = 3x$$


$$w \in L_1 \rightarrow V(w) = 3x + 1$$


$$w \in L_2 \rightarrow V(w) = 3x + 2$$


$$w = \varepsilon$$


$$V(w0) = V$$

$$V(w1) = V$$

PROBLEM 4

Suppose M is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

Suppose M_2 is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{p_0, p_1, p_2, p_3, p_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

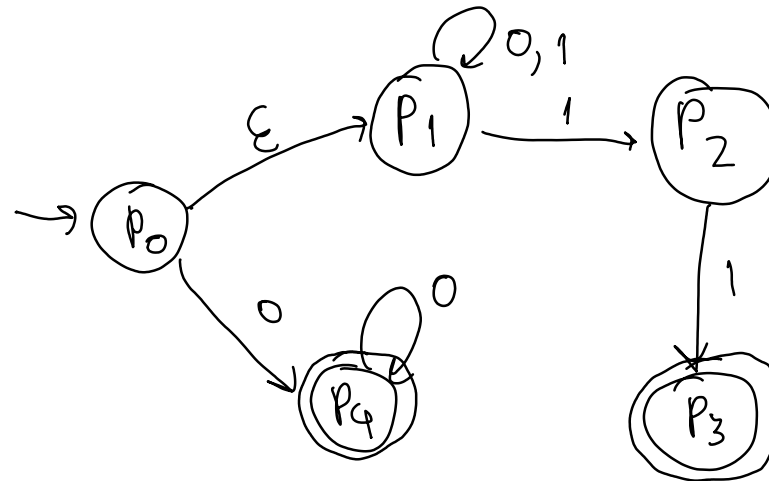
	ϵ	0	1
p_0	$\{p_1\}$	$\{p_4\}$	$\{\}$
p_1	$\{\}$	$\{p_1\}$	$\{p_1, p_2\}$
p_2	$\{\}$	$\{\}$	$\{p_3\}$
p_3	$\{\}$	$\{\}$	$\{\}$
p_4	$\{\}$	$\{p_4\}$	$\{\}$

- p_0 is the initial state,
- $F = \{p_3, p_4\}$

4.1) Draw a state diagram of machine M_2 .

4.2) Write a computation tree of M_2 on the string 01011. Is this string accepted by M_2 ?

4.3) Apply subset construction to construct a (deterministic) finite automaton that is equivalent to M_2 .

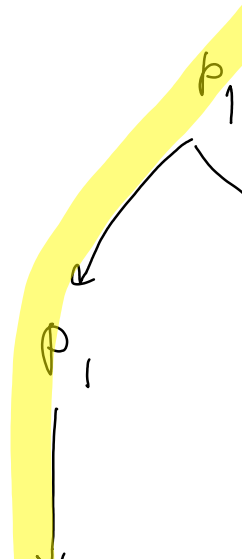


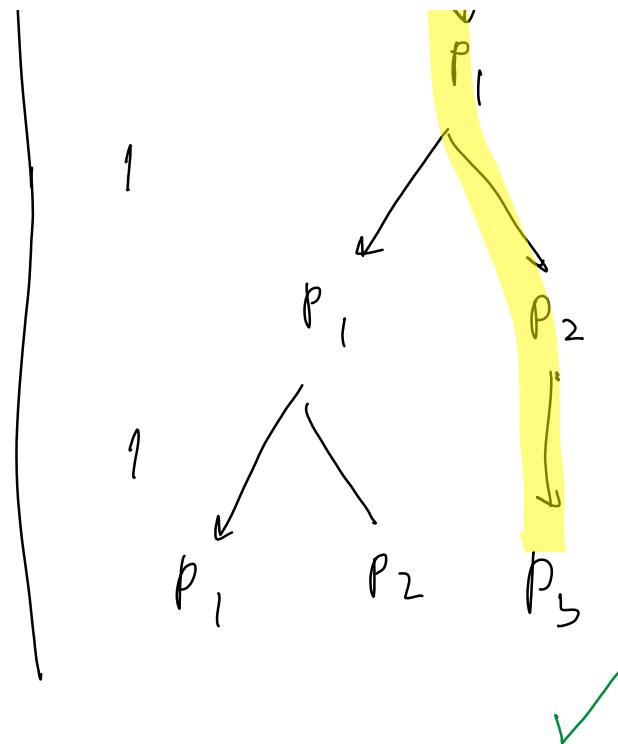
01011

0

1

0





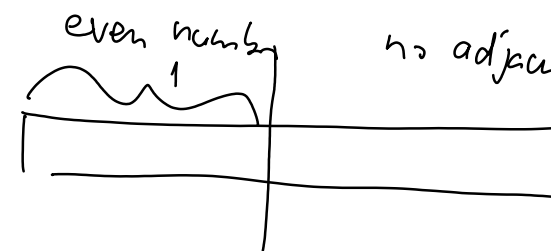
$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$

$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$

$\overline{L_1 L_2}$

$\varepsilon \in L_1$
 $\varepsilon \in L_2$

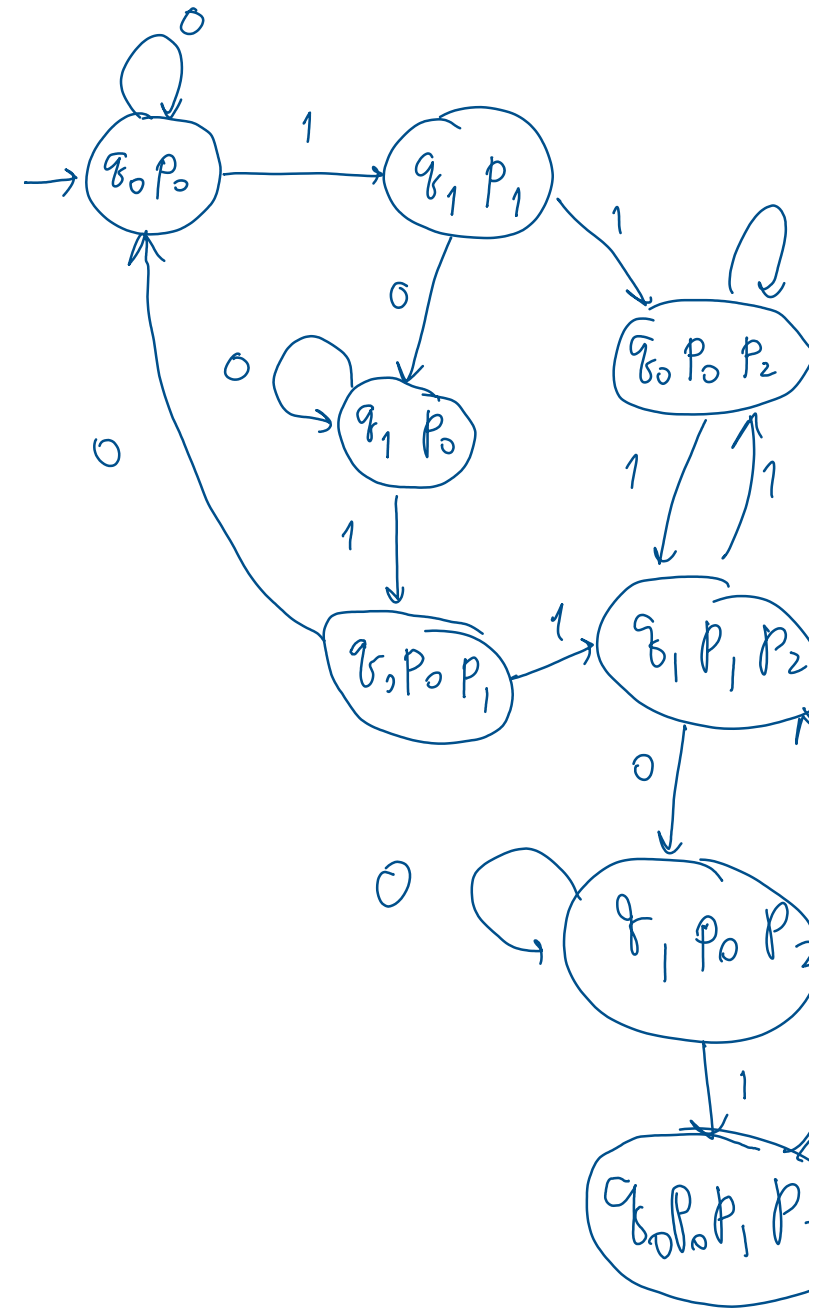
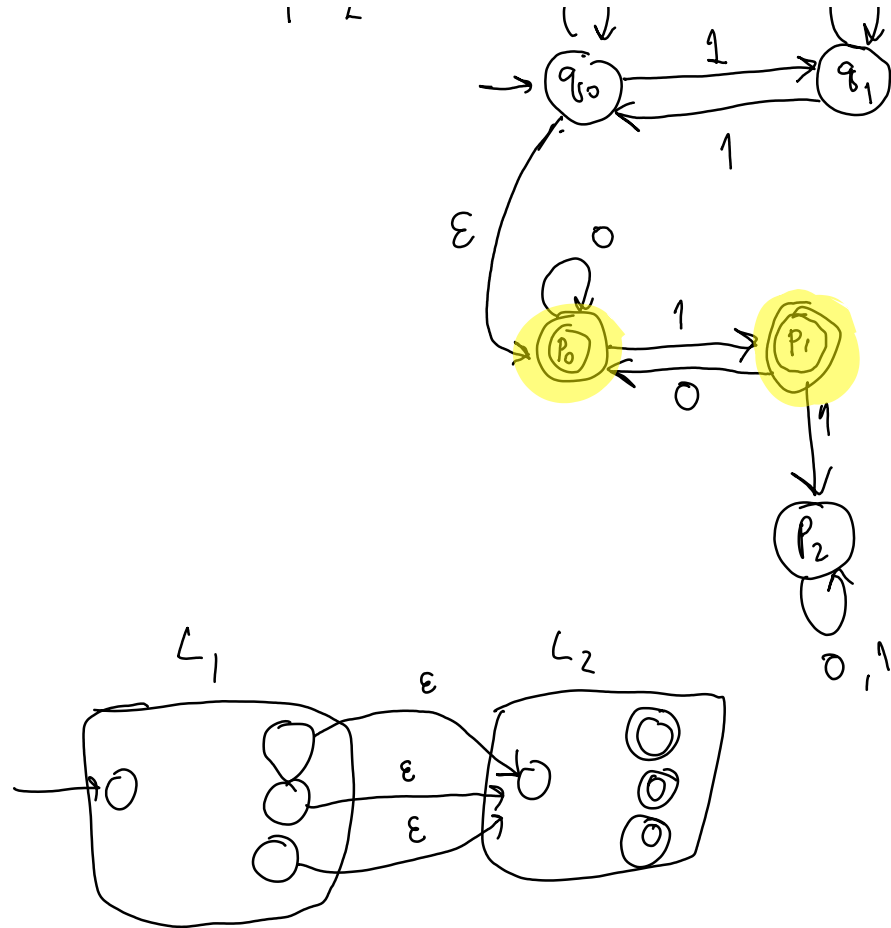
$L_1 L_2$



M, M_n

$\cap^0$

$\cap^0$



Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

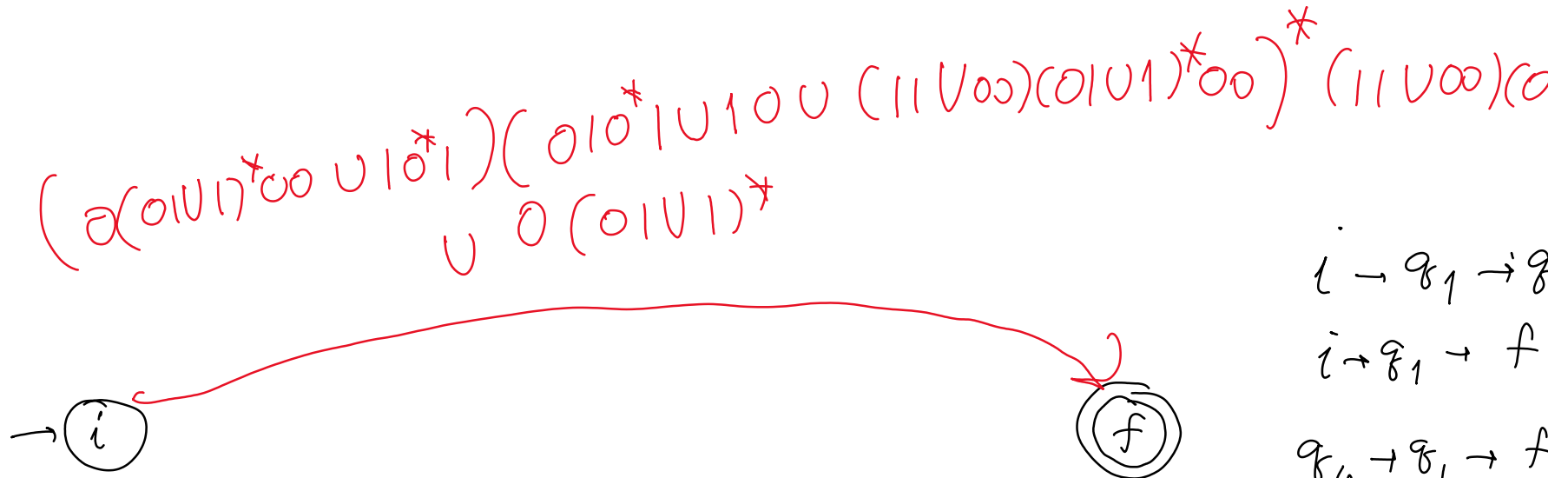
- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	0	1
q_0		
q_1		
q_2		
q_3		
q_4		

q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_4
q_4	q_0	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.



PROBLEM 2



Let β be the following regular expression over the alphabet $\Sigma = \{0,1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

Show how to construct a nondeterministic finite automaton N equivalent to this regular express $L(N) = L(\beta)$.

